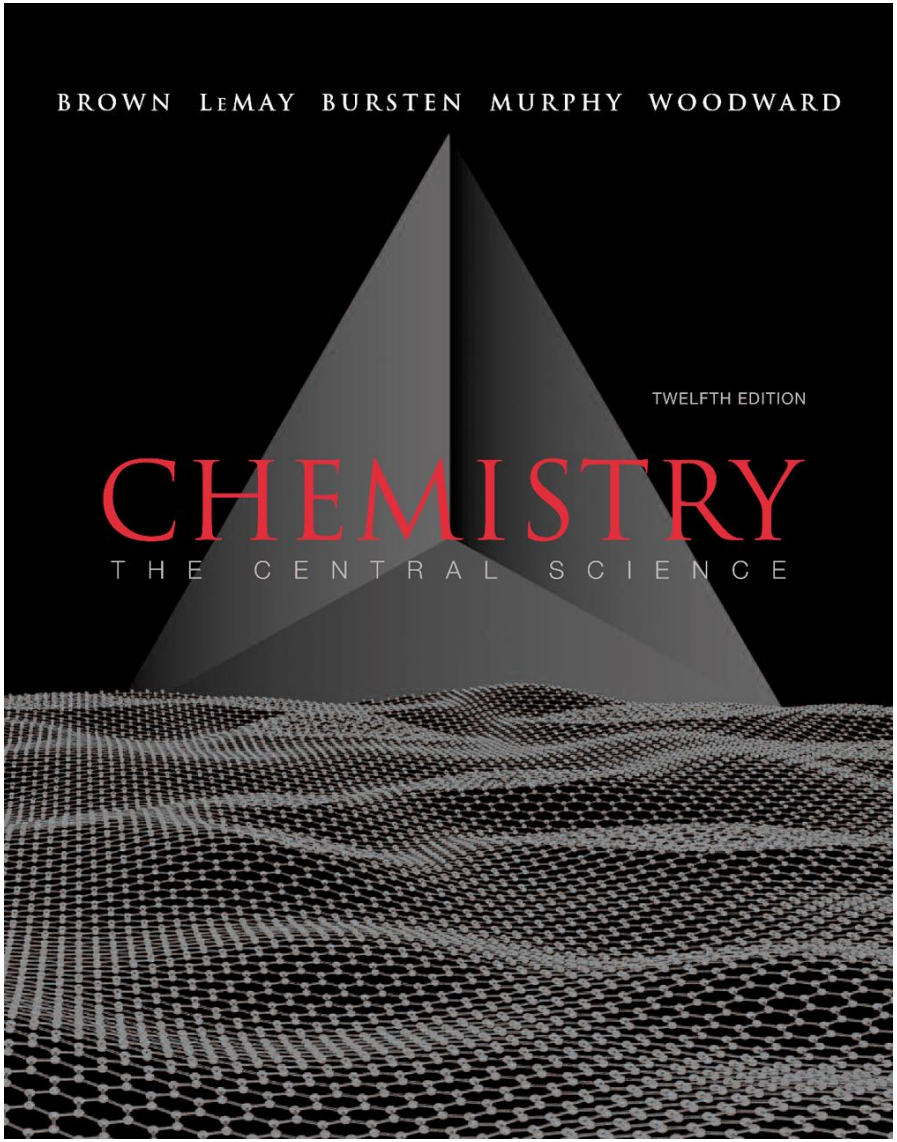


Lecture Presentation

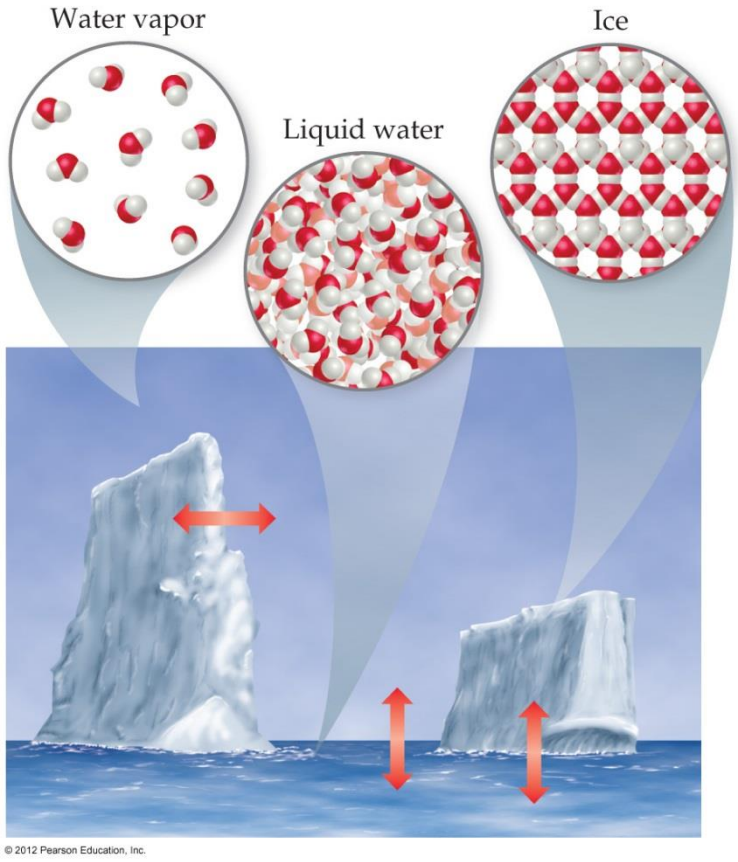
Chapter 1

Introduction: Matter and Measurement

John D. Bookstaver
St. Charles Community College
Cottleville, MO

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Chemistry



In this science we study **matter**, its **properties**, and its **behavior**.

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Why Study Chemistry?

Chemistry lies near the heart of many matters of public concern, such as improvement of health care, conservation of natural resources, protection of the environment, and the supply of energy needed to keep society running. Using chemistry, we have discovered and continually improved upon pharmaceuticals, fertilizers and pesticides, plastics, solar panels, LEDs, and building materials.

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Energy

Solar panels are composed of specially treated silicon.



Biochemistry

The flash of the firefly results from a chemical reaction in the insect.



Chemistry

Technology

LED's (light emitting diodes) are formed from elements such as gallium, arsenic and phosphorus.



Medicine

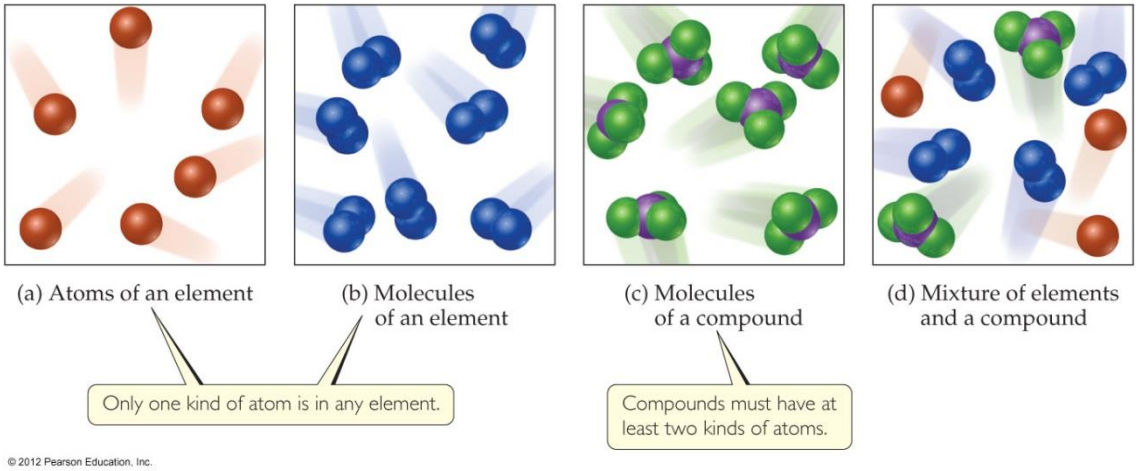
Connectors and tubing for medical procedures such a intravenous injections are made from plastics highly resistant to chemical attack



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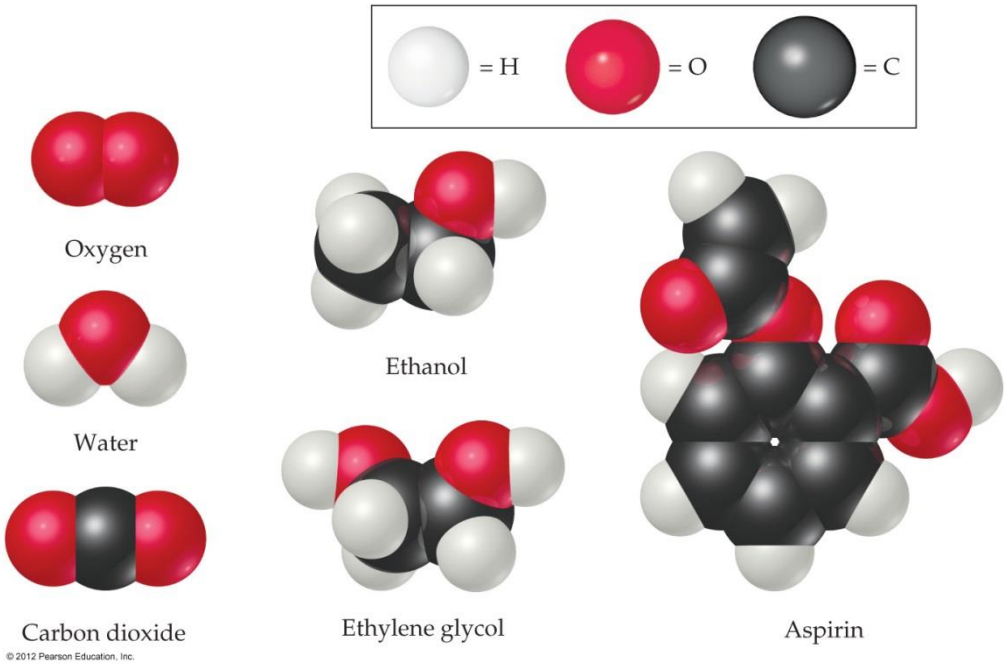
Matter

We define **matter** as anything that has mass and takes up space.



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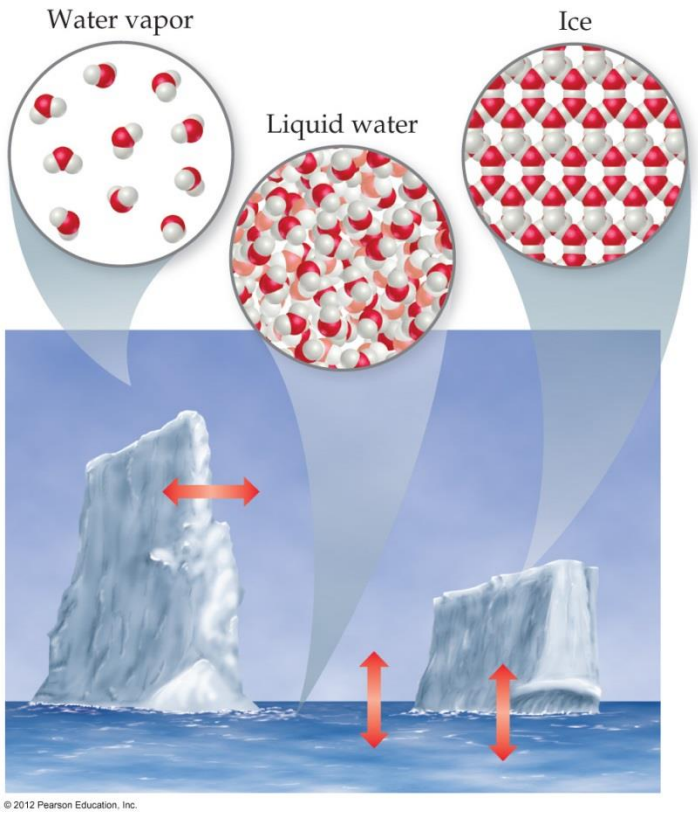
Matter



- **Atoms** are the building blocks of matter.
- Each **element** is made of the same kind of atom.
- A **compound** is made of two or more different kinds of elements.

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States of Matter



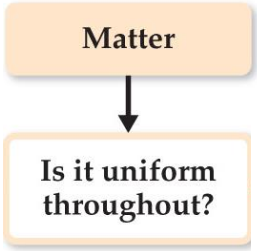
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Classification of Matter



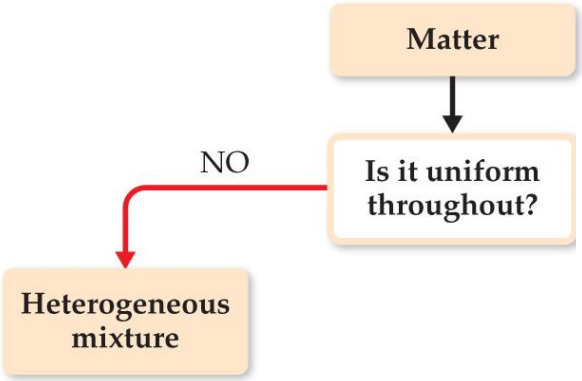
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Classification of Matter



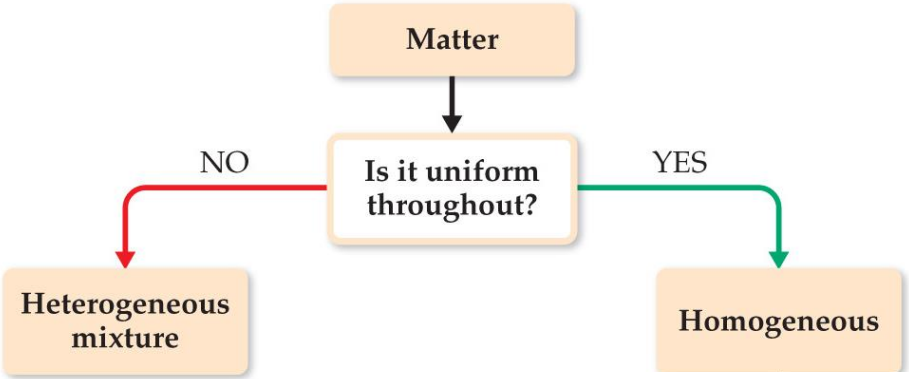
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Classification of Matter



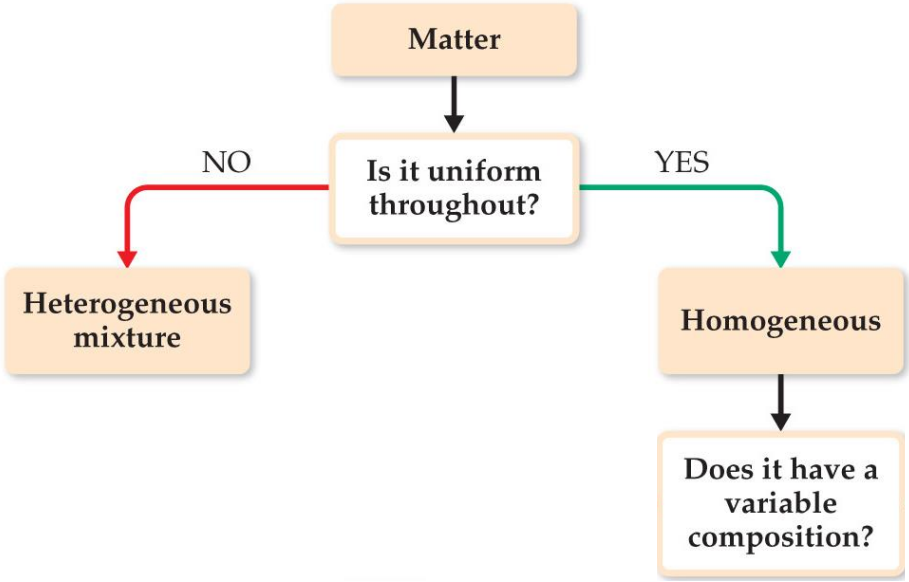
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Classification of Matter



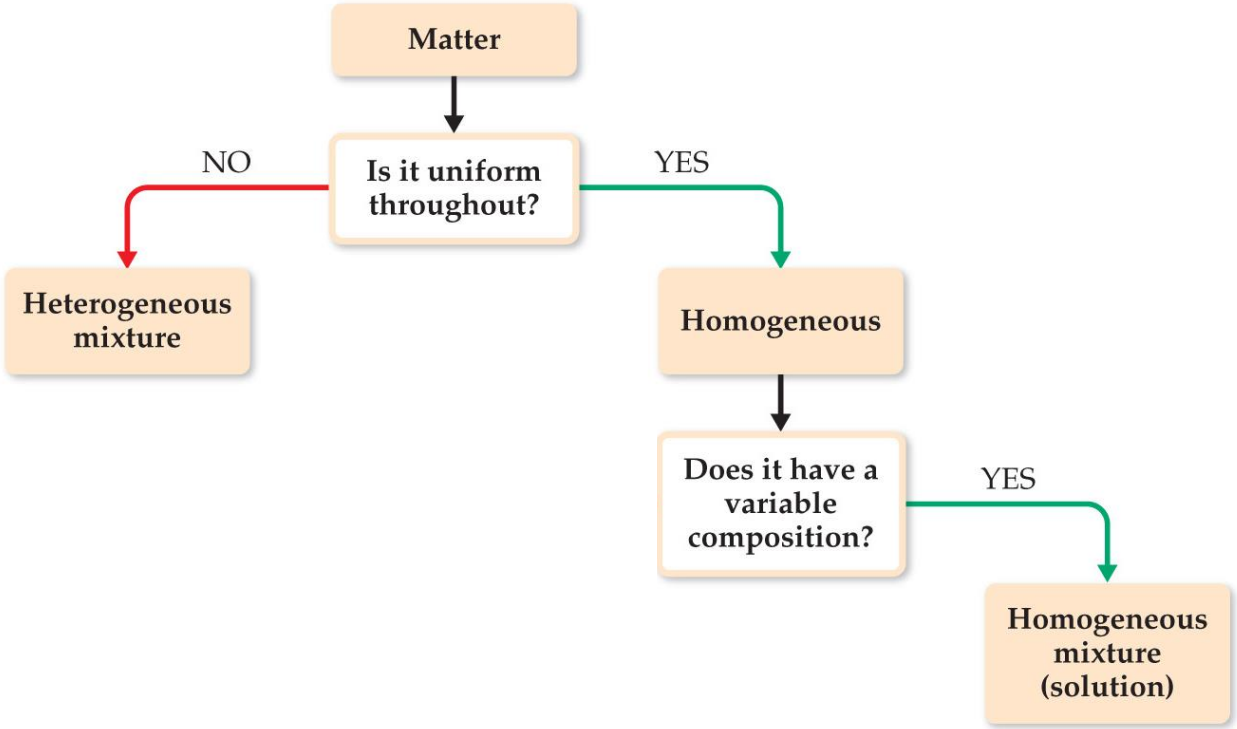
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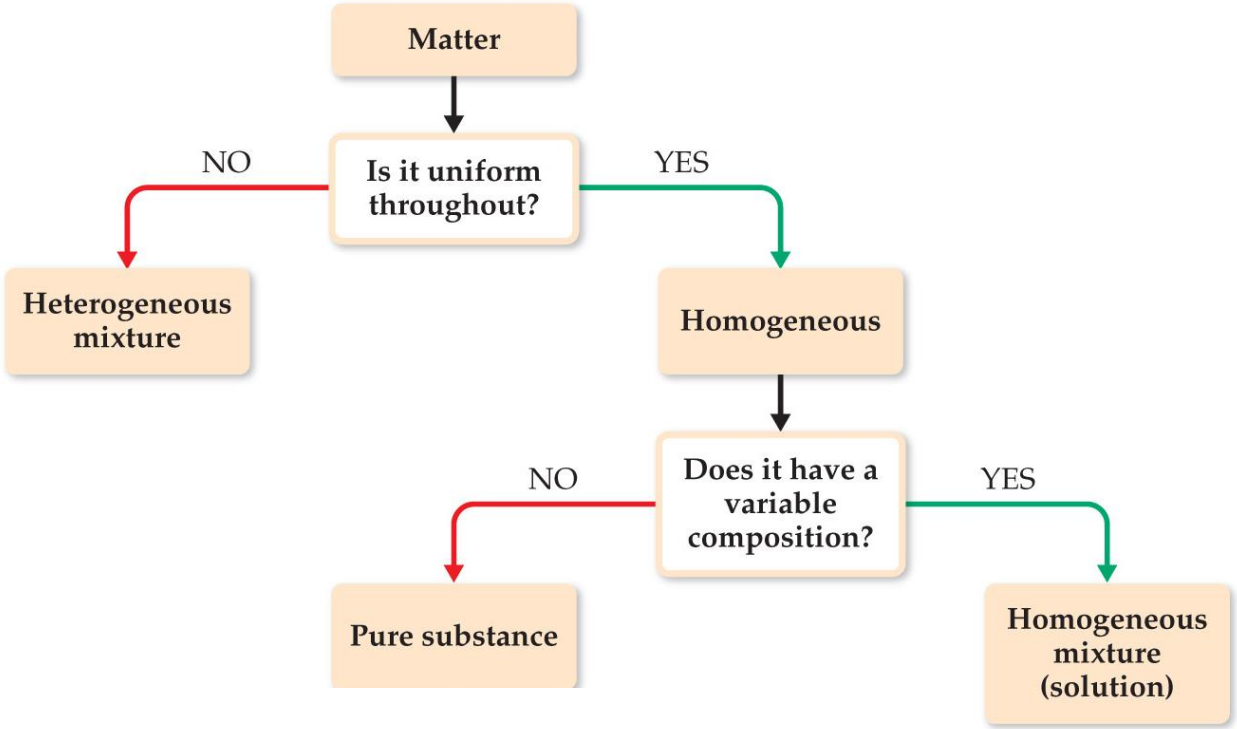
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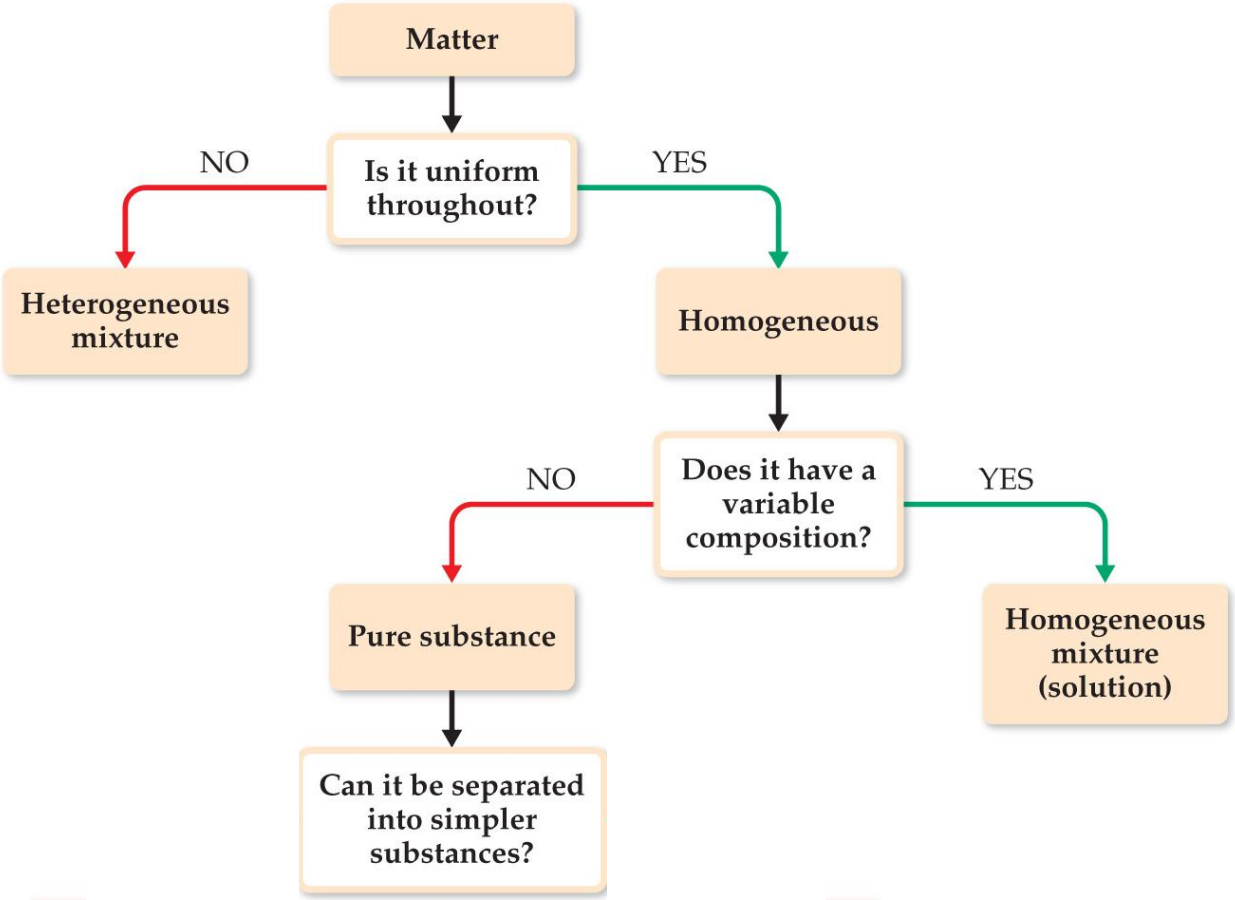
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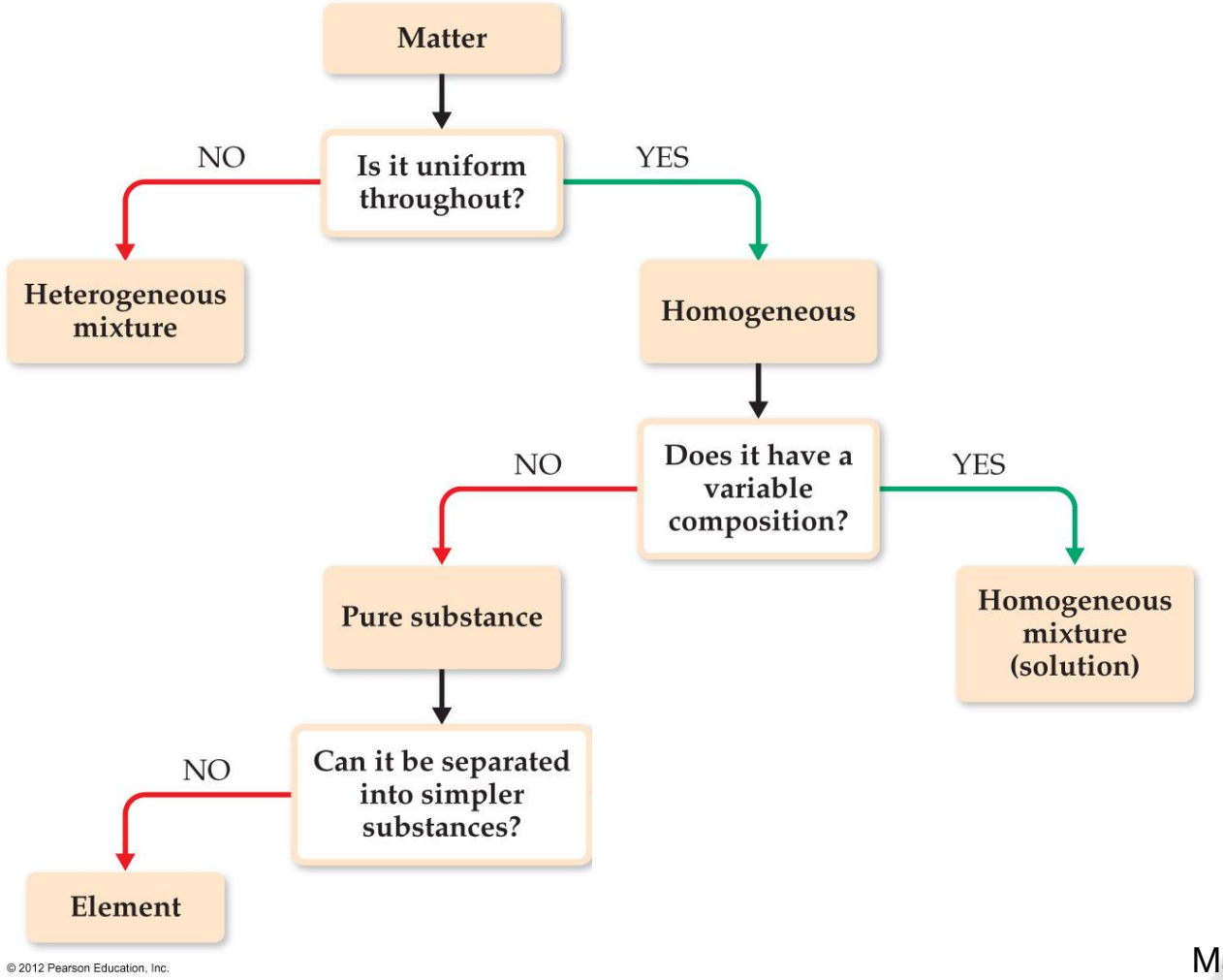
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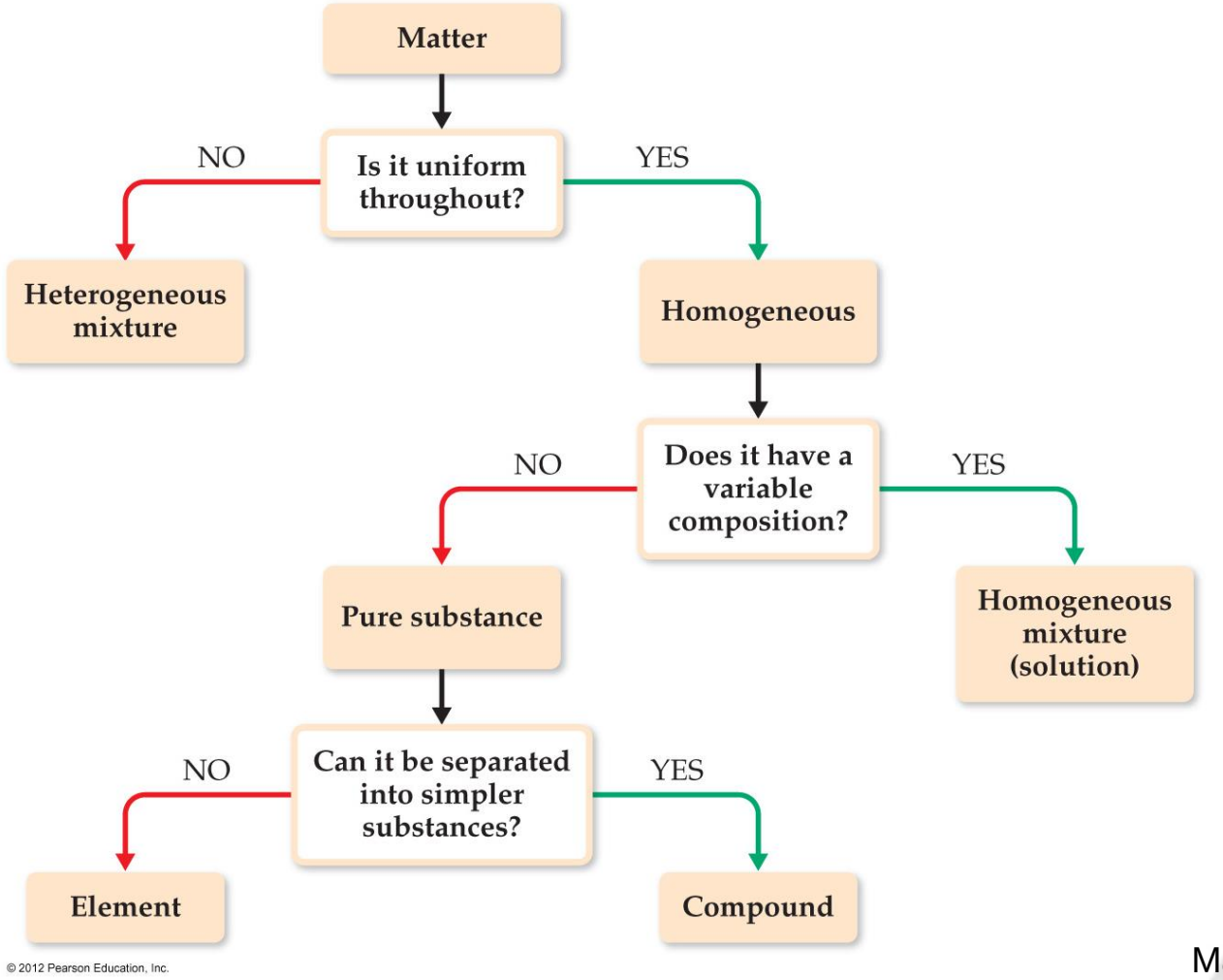
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Classification of Matter



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Classification of Matter



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Properties and Changes of Matter

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Types of Properties

- Physical Properties…
  - Can be observed without changing a substance into another substance.
    - Boiling point, density, mass, volume, etc.
- Chemical Properties…
  - Can *only* be observed when a substance is changed into another substance.
    - Flammability, corrosiveness, reactivity with acid, etc.

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Types of Properties

- Intensive Properties…
  - Are independent of the amount of the substance that is present.
    - Density, boiling point, color, etc.
- Extensive Properties…
  - Depend upon the amount of the substance present.
    - Mass, volume, energy, etc.

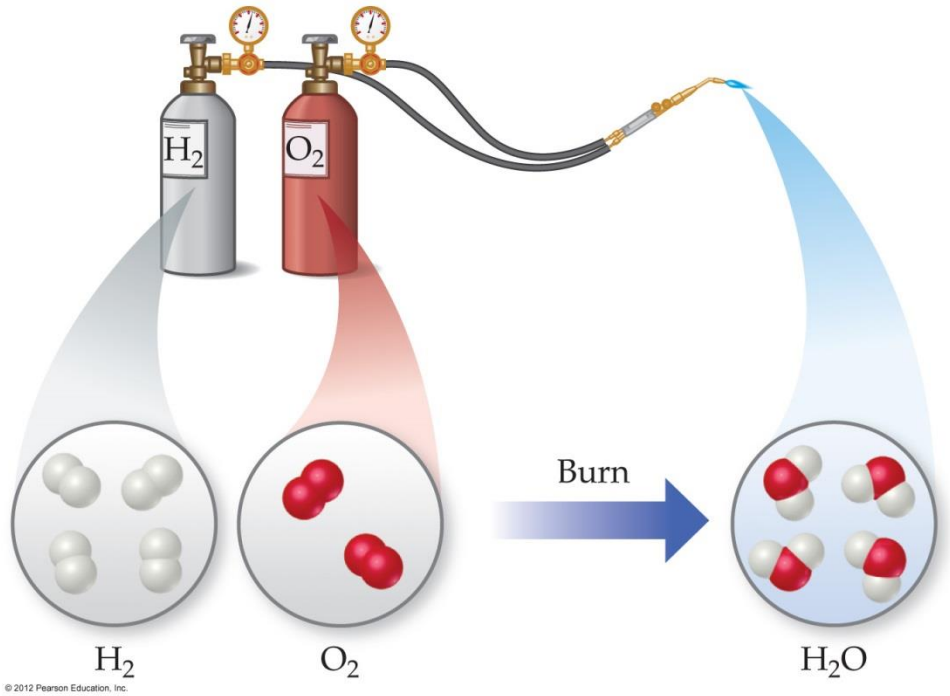
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Types of Changes

- Physical Changes
  - These are changes in matter that do not change the composition of a substance.
    - Changes of state, temperature, volume, etc.
- Chemical Changes
  - Chemical changes result in new substances.
    - Combustion, oxidation, decomposition, etc.

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Chemical Reactions



In the course of a chemical reaction, the reacting substances are converted to new substances.

Matter And Measurement

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Separation of Mixtures

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Filtration

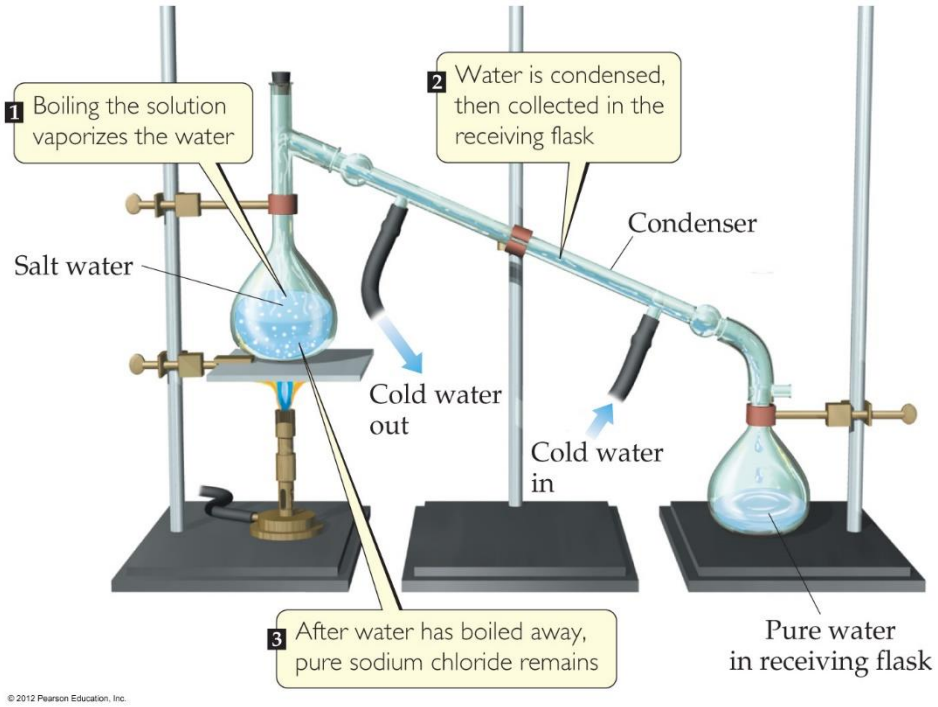




In filtration, solid substances are separated from liquids and solutions.

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Distillation

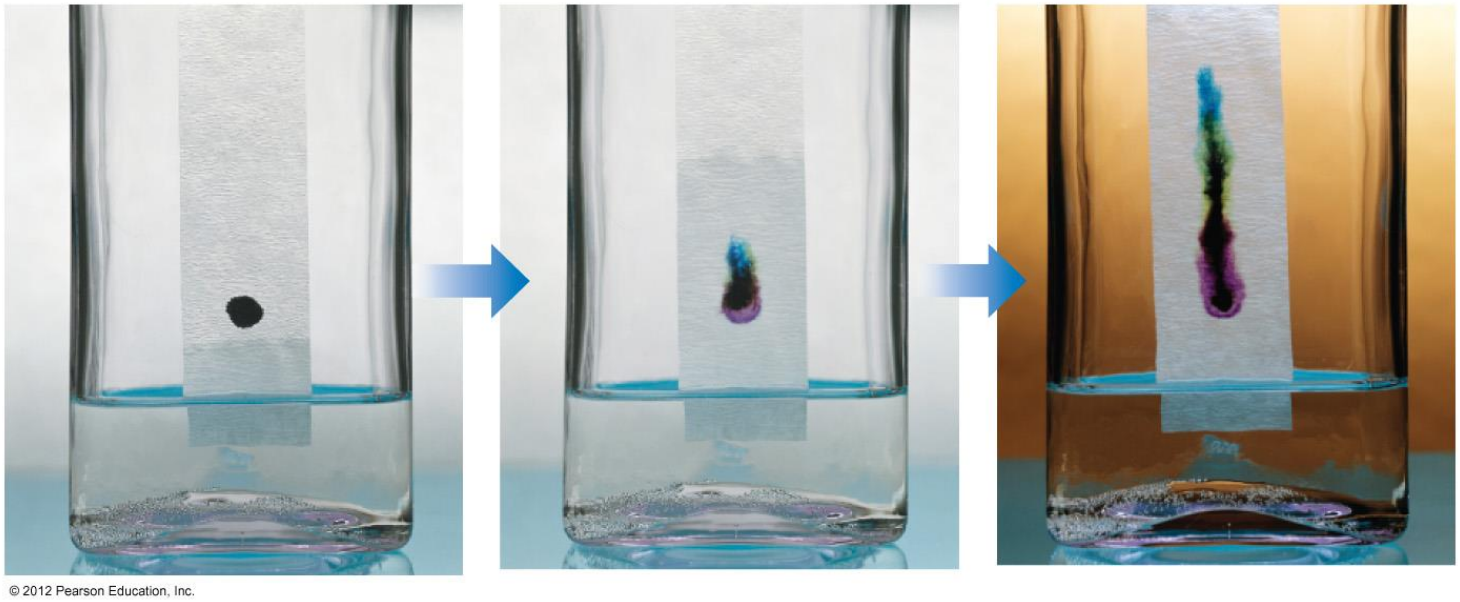


Distillation uses differences in the boiling points of substances to separate a homogeneous mixture into its components.

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Chromatography

This technique separates substances on the basis of differences in solubility in a solvent.



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Units of Measurement

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SI Units

TABLE 1.4 • SI Base Units

Physical Quantity	Name of Unit	Abbreviation
Mass	Kilogram	kg
Length	Meter	m
Time	Second	s or sec
Temperature	Kelvin	K
Amount of substance	Mole	mol
Electric current	Ampere	A or amp
Luminous intensity	Candela	cd

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- *Système International d'Unités*
- A different base unit is used for each quantity.

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Metric System

Prefixes convert the base units into units that are appropriate for the item being measured.

TABLE 1.5 • Prefixes Used in the Metric System and with SI Units

Prefix	Abbreviation	Meaning	Example
Peta	P	10^{15}	1 petawatt (PW) $= 1 \times 10^{15}$ watts[a]
Tera	T	10^{12}	1 terawatt (TW) $= 1 \times 10^{12}$ watts
Giga	G	10^{9}	1 gigawatt (GW) $= 1 \times 10^{9}$ watts
Mega	M	10^{6}	1 megawatt (MW) $= 1 \times 10^{6}$ watts
Kilo	k	10^{3}	1 kilowatt (kW) $= 1 \times 10^{3}$ watts
Deci	d	10^{-1}	1 deciwatt (dW) $= 1 \times 10^{-1}$ watt
Centi	c	10^{-2}	1 centiwatt (cW) $= 1 \times 10^{-2}$ watt
Milli	m	10^{-3}	1 milliwatt (mW) $= 1 \times 10^{-3}$ watt
Micro	μ[b]	10^{-6}	1 microwatt (μW) $= 1 \times 10^{-6}$ watt
Nano	n	10^{-9}	1 nanowatt (nW) $= 1 \times 10^{-9}$ watt
Pico	p	10^{-12}	1 picowatt (pW) $= 1 \times 10^{-12}$ watt
Femto	f	10^{-15}	1 femtowatt (fW) $= 1 \times 10^{-15}$ watt
Atto	a	10^{-18}	1 attowatt (aW) $= 1 \times 10^{-18}$ watt
Zepto	z	10^{-21}	1 zeptowatt (zW) $= 1 \times 10^{-21}$ watt

[a]The watt (W) is the SI unit of power, which is the rate at which energy is either generated or consumed. The SI unit of energy is the joule (J); 1 J = 1 kg · m^2/s^2 and 1 W=1 J/s.

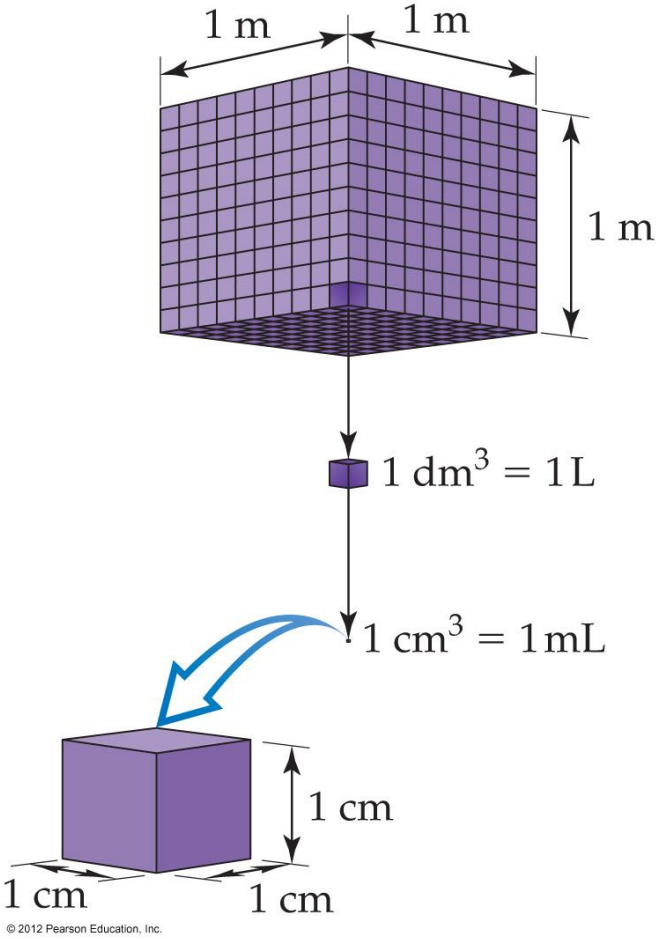
[b]Greek letter mu, pronounced "mew."

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Volume

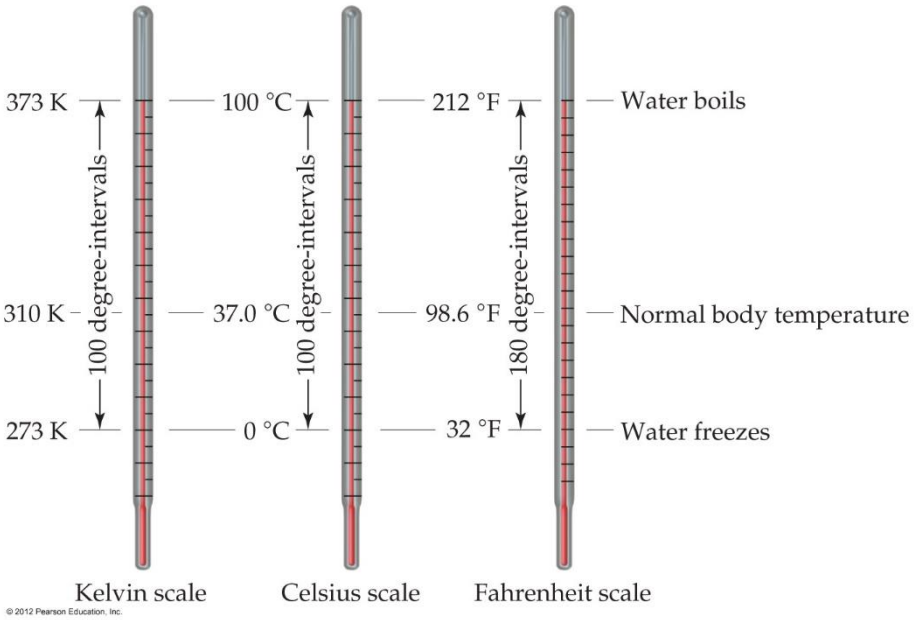
- The most commonly used metric units for volume are the liter (L) and the milliliter (mL).
  - A liter is a cube 1 decimeter (dm) long on each side.
  - A milliliter is a cube 1 centimeter (cm) long on each side.



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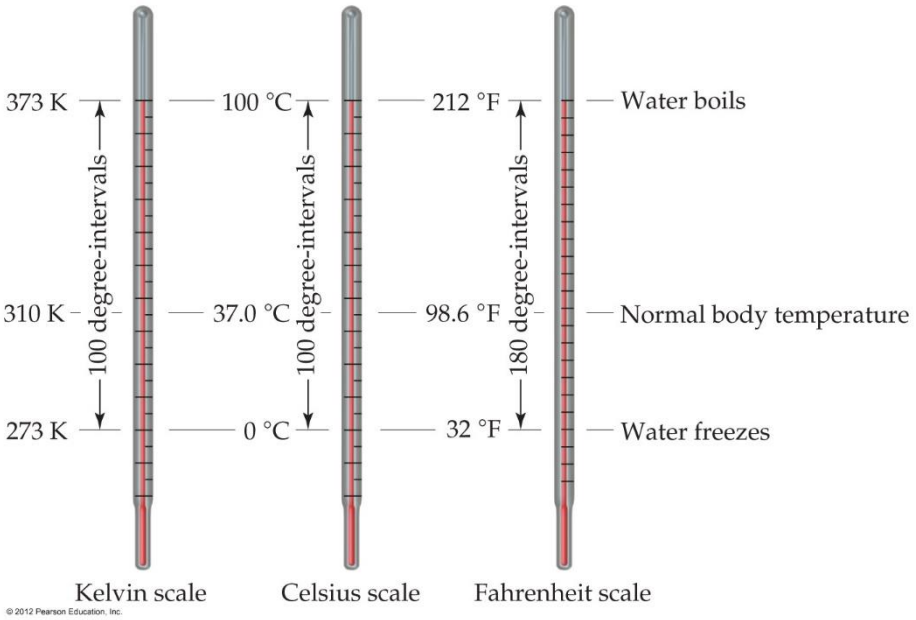
Temperature



By definition **temperature** is a measure of the average kinetic energy of the particles in a sample.

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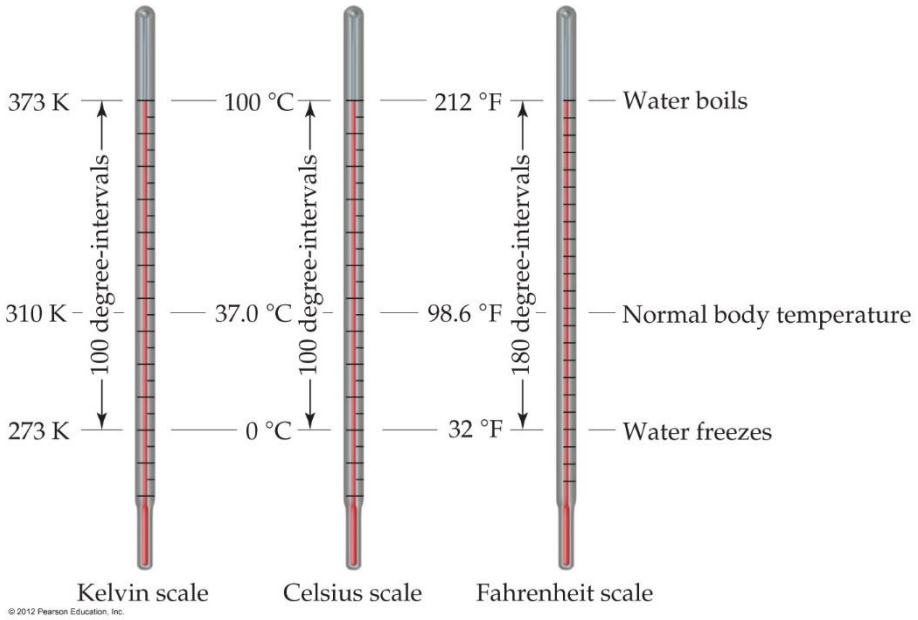
Temperature



- In scientific measurements, the Celsius and Kelvin scales are most often used.
- The Celsius scale is based on the properties of water.
  - 0 °C is the freezing point of water.
  - 100 °C is the boiling point of water.

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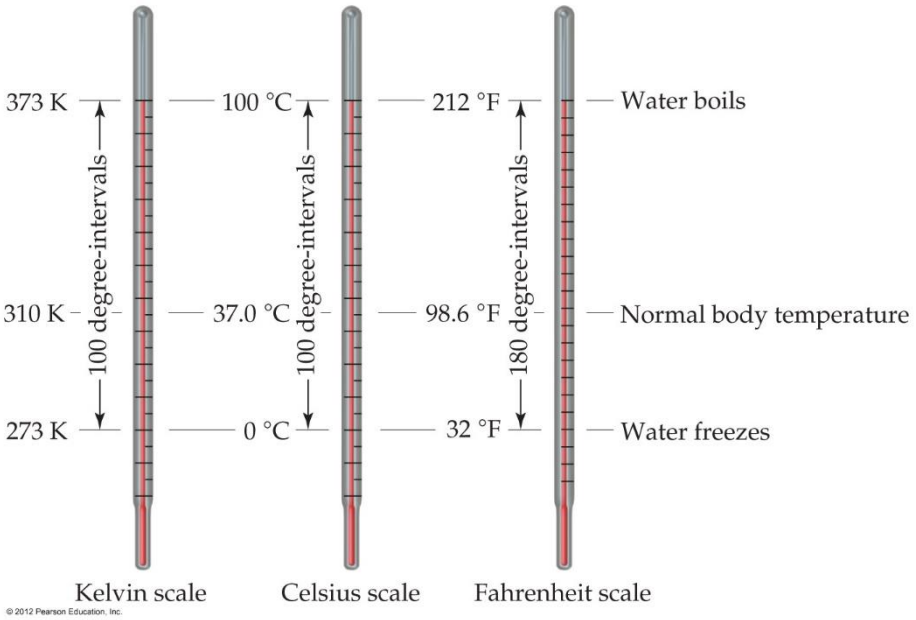
Temperature



- The kelvin is the SI unit of temperature.
- It is based on the properties of gases.
- There are no negative Kelvin temperatures.
- K = °C + 273.15

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Temperature



- The Fahrenheit scale is not used in scientific measurements.
- °F = 9/5(°C) + 32
- °C = 5/9(°F − 32)

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Derived Units

- Density is a physical property of a substance.
- It has units (*g*/*mL*, for example) that are derived from the units for mass and volume.

$$d = \frac{m}{V}$$

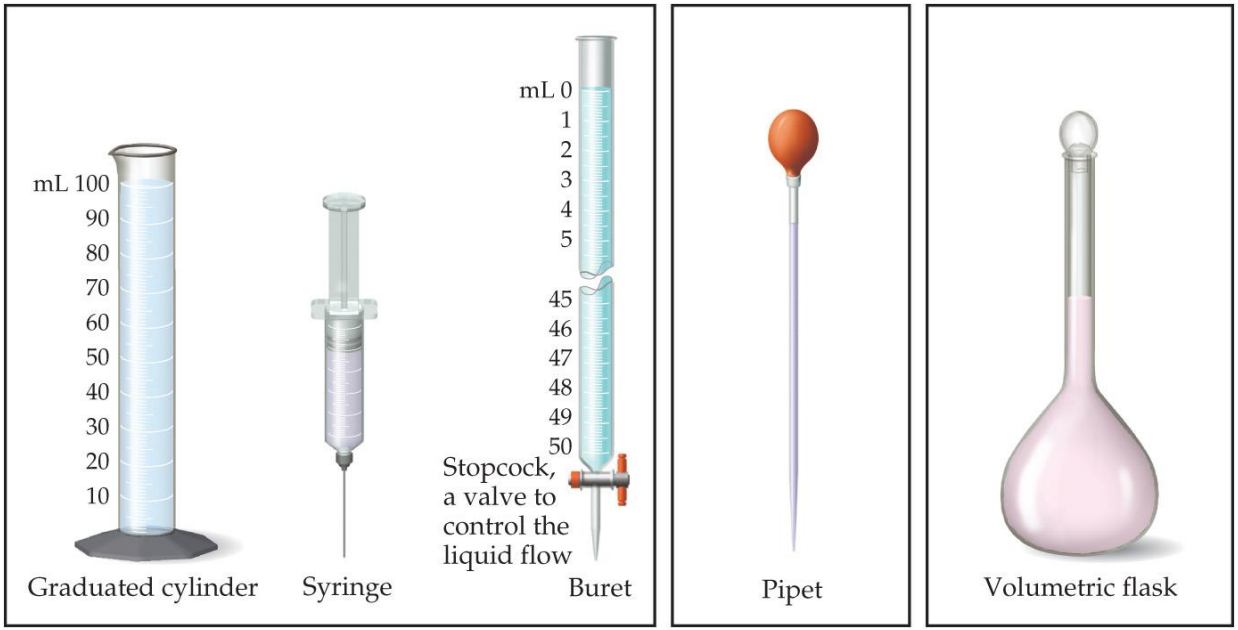
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Uncertainty in Measurement

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Uncertainty in Measurements

Different measuring devices have different uses and different degrees of accuracy.



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Significant Figures

- The term **significant figures** refers to digits that were measured.
- When rounding calculated numbers, we pay attention to significant figures so we do not overstate the accuracy of our answers.

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Significant Figures

1. All nonzero digits are significant.
2. Zeroes between two significant figures are themselves significant.
3. Zeroes at the beginning of a number are never significant.
4. Zeroes at the end of a number are significant if a decimal point is written in the number.

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Significant Figures

- When addition or subtraction is performed, answers are rounded to the least significant **decimal place**.
- When multiplication or division is performed, answers are rounded to the number of digits that corresponds to the ***least* number of significant figures** in any of the numbers used in the calculation.

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Toplama ve çıkarma sonucu, ondalık basamak sayısı *en az* olan nicelikteki ile aynı sayıda ondalık basamakla ifade edilmelidir.

Aşağıdaki kütlelerin toplamını ele alalım:

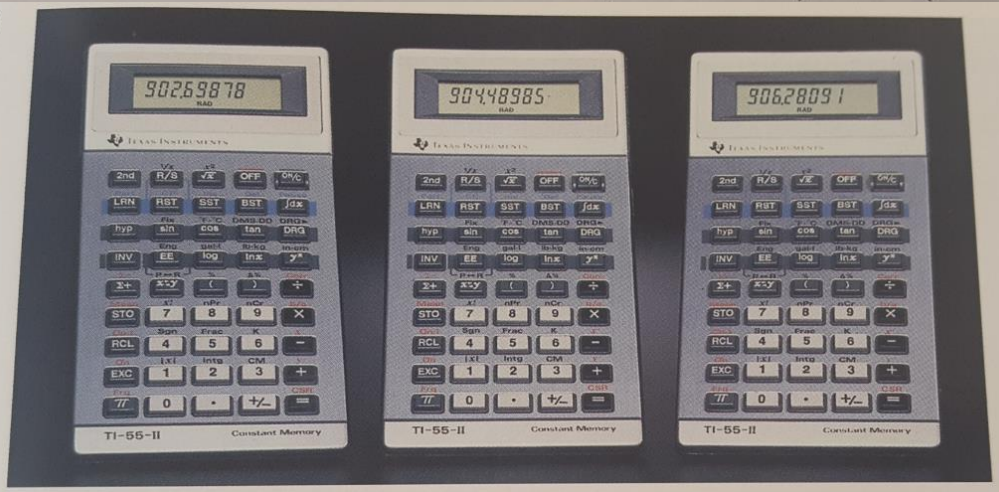
$$\begin{array}{r} 15{,}02\ \text{g} \\ 9986{,}0\ \text{g} \\ 3{,}518\ \text{g} \\ \hline 10\,004{,}538\ \text{g} \end{array}$$

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Çarpma ve bölmenin sonucu ancak hesaplamada kesinliği en az bilinen niceliğinki kadar anlamlı rakam içerebilir.

Aşağıdaki çarpma zincirinde, dikdörtgen şeklindeki bir odun kütüğünün hacmini bulmak için sonucu *üç* anlamlı rakama yuvarlamalıyız. Şekil 1-12 bunu anlamanıza yardım eder.

$$\underset{\text{(4 an. rak.)}}{14{,}79\ \text{cm}} \times \underset{\text{(4 an. rak.)}}{12{,}11\ \text{cm}} \times \underset{\text{(3 an. rak.)}}{5{,}05\ \text{cm}} = \underset{\text{(3 an. rak.)}}{904\ \text{cm}^3}$$



▲ ŞEKİL 1-12
Çarpmada anlamlı rakam kuralı
14,79 cm × 12,11 cm × 5,05 cm işleminde kesinliği en az olan sayı 5,05 cm'dir. Hesap makinelerinde 14,79 ve 12,11'in sırasıyla 5,04, 5,05 ve 5,06 ile çarpımları gösterilmiştir. Üç sonuçta da yalnızca ilk iki rakam 90… aynıdır, üçüncü rakamda değişiklikler başlar. Rakamları üçüncünün ötesine götürmekte haklı bir hedefimiz olamaz. Hacmi 904 cm^3 olarak ifade ederiz. Buradaki gibi ayrıntılı bir analiz yerine genellikle daha basit bir kavram kullanırız: Bir çarpmanın sonucu ancak en az kesinlikle bilinen niceliğin içerdiği kadar anlamlı rakam içerebilir.

Accuracy versus Precision

- **Accuracy** refers to the proximity of a measurement to the true value of a quantity.
- **Precision** refers to the proximity of several measurements to each other.



Good accuracy
Good precision



Poor accuracy
Good precision



Poor accuracy
Poor precision

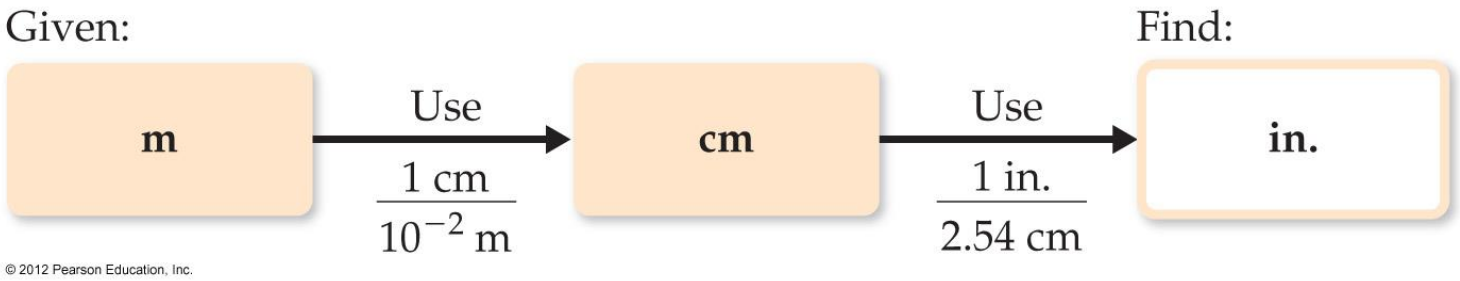
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Dimensional Analysis

- We use **dimensional analysis** to convert one quantity to another.
- Most commonly, dimensional analysis utilizes **conversion factors** (e.g., 1 in. = 2.54 cm)

$$\frac{1\text{ in.}}{2.54\text{ cm}} \quad \text{or} \quad \frac{2.54\text{ cm}}{1\text{ in.}}$$



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Dimensional Analysis

Use the form of the conversion factor that puts the sought-for unit in the numerator:

$$\cancel{\text{Given unit}} \times \frac{\text{desired unit}}{\cancel{\text{given unit}}} = \text{desired unit}$$

Conversion factor

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Dimensional Analysis

- For example, to convert 8.00 m to inches,
  - convert m to cm
  - convert cm to in.

$$8.00\ \cancel{m} \times \frac{100\ \cancel{cm}}{1\ \cancel{m}} \times \frac{1\ \text{in.}}{2.54\ \cancel{cm}} = 315\ \text{in.}$$

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Earth's oceans contain approximately $1.36 \times 10^9\ \text{km}^3$ of water. Calculate the volume in liters.

$$\left(\frac{10^3\ \text{m}}{1\ \text{km}}\right)^3 = \frac{10^9\ \text{m}^3}{1\ \text{km}^3}$$

$$1\ \text{L} = 10^{-3}\ \text{m}^3$$

$$\text{Volume in liters} = (1.36 \times 10^9\ \cancel{\text{km}^3})\left(\frac{10^9\ \cancel{\text{m}^3}}{1\ \cancel{\text{km}^3}}\right)\left(\frac{1\ \text{L}}{10^{-3}\ \cancel{\text{m}^3}}\right) = 1.36 \times 10^{21}\ \text{L}$$

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